

A TIGHT FAMILY FOR THE STATE COMPLEXITY OF LINEAR SUBSEQUENCES OF AUTOMATIC SEQUENCES

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ABSTRACT. Open Problem 1 of Moradi, Rampersad and Shallit (arXiv:2512.10017v5, §4) asks for a class of automatic sequences $(h_m(i))_{i \geq 0}$, each generated by an m -state msd-first DFAO, whose linear subsequences $(h_m(ni))_{i \geq 0}$ have state complexity at least Cnm^2 for infinitely many n and some constant C . We answer it with the generalised Thue–Morse family $h_m(i) = s_2(i) \bmod m$ (s_2 the binary digit sum): for every $m \geq 2$ and odd $n \geq 3$ the minimal msd-first DFAO of $(h_m(ni))_{i \geq 0}$ has exactly $m^2(n-1) - m \max(2^{p-1}, r)$ states, where $n-1 = 2^p + r$, $0 < r \leq 2^p$. This is at least $\frac{1}{2}nm^2$ for every odd $n \geq 3$, and $\limsup_n \text{sc}(h_m(ni))/(nm^2)$ equals $1 - 1/(3m) \geq 5/6$; the value $C = 1/2$ is attained exactly at $(m, n) = (2, 3)$, so within this family it is not improvable uniformly over odd n . The proof joins three ingredients: a Myhill–Nerode lemma for every digit-additive sequence and every n , generalising the authors’ own optimality theorem; a reachability lemma for odd n from Gelfond’s 1968 equidistribution theorem, replacing their Thue–Morse-specific one; and the known factor complexity of the generalised Thue–Morse word.

1. BACKGROUND AND THE PROBLEM

Fix a base $k \geq 2$ and write $\Sigma_k = \{0, \dots, k-1\}$; for $x \in \Sigma_k^*$ let $[x]_k$ be the integer it represents, most-significant digit first, and for $i \in \mathbb{N}$ let $(i)_k$ be its leading-zero-free representation. A sequence $h = (h(i))_{i \geq 0}$ over a finite alphabet Δ is k -automatic if $h(i) = \tau(\delta(q_0, (i)_k))$ for some DFAO $M = (Q, \Sigma_k, \delta, q_0, \Delta, \tau)$. Input is msd-first with leading zeros allowed throughout, so we assume M is *zero-consistent*: $\delta(q_0, 0) = q_0$, which is exactly what makes the recursion $h'(ki+b) = \delta(h'(i), b)$ valid at $i = 0$ as well. The *interior sequence* is $h'(i) = \delta(q_0, (i)_k) \in Q$, generated by the *intrinsic* DFAO $\widehat{M} = (Q, \Sigma_k, \delta, q_0, Q, \text{id})$. We write $\rho_x(n)$ for the number of length- n factors of x and $\text{sc}(x)$ for the number of states of the minimal msd-first DFAO generating x .

Moradi, Rampersad and Shallit [6] relate these two quantities. For $n \geq 1$ and $0 \leq c < n$, their Theorem 9(a) generates $(h(ni+c))_{i \geq 0}$ by a DFAO whose states are length- n factors of h' , the state reached on input x being the window $[h'(n[x]_k), \dots, h'(n[x]_k + n-1)]$ with output $[r_0, \dots, r_{n-1}] \mapsto \tau(r_c)$. Hence $\text{sc}(h(ni+c)) \leq \rho_{h'}(n)$; with $\rho_x(n) \leq kn \rho_{\widehat{M}}(2) \leq knm^2$ [1, Thm. 10.3.1] this gives their Corollary 10(a): $\text{sc}(h(ni+c)) \leq knm^2$. They then write [6, §4]:

We do not know if the bounds in Corollary 10 are tight. This seems worthy of further study, so we state it as an open problem.

Open Problem 1 ([6, §4]). Find a class of automatic sequences $(h_m(i))_{i \geq 0}$, generated by a DFAO of m states with msd-first input, such that there is a constant C so the linear subsequences $(h_m(ni))_{i \geq 0}$ have at least the state complexity Cnm^2 for infinitely many n .

(Quoted verbatim from arXiv:2512.10017v5, 2 April 2026. It appears again, verbatim, as Problem 1 of Moradi’s MMath thesis [5], whose “More Open Problems” section leaves it unresolved, so the problem is open as of its author’s latest treatment.)

For $m = 2$ the authors settle the state count: their Theorem 16 shows the Theorem 9 construction optimal for the Thue–Morse word t and odd n ; their Lemma 15, quoting [2], shows every length- n factor of t occurs at a position divisible by odd n ; and with Brlek’s factor complexity of t [3] this yields $\text{sc}(t(ni)) = \rho_t(n/\nu_2(n))$ (their Theorem 14) and the explicit Corollary 17. What is missing is the m -dependence — a family in which the m^2 of Corollary 10 is visible.

2. THE FAMILY

For $m \geq 2$ let $s_2(i)$ be the number of 1s in the binary representation of i , and let

$$t_{2,m} = (s_2(i) \bmod m)_{i \geq 0}$$

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be the *generalised Thue–Morse word* over $\mathbb{Z}/m\mathbb{Z}$; for $m = 2$ this is the Thue–Morse word. It is generated by the m -state zero-consistent msd-first DFAO

$$M_m = (\mathbb{Z}/m\mathbb{Z}, \{0, 1\}, \delta, 0, \mathbb{Z}/m\mathbb{Z}, \text{id}), \quad \delta(q, d) = q + d \bmod m,$$

equivalently by the 2-uniform morphism $j \mapsto j(j+1)$ on $\mathbb{Z}/m\mathbb{Z}$ with identity coding. State q is reached by 1^q and the coding is injective, so distinct states are separated by the empty word: M_m is minimal, $\text{sc}(t_{2,m}) = m$, and the interior sequence of M_m is $t_{2,m}$ itself. So $t_{2,m}$ is a legitimate candidate for Problem 1. The one structural property we use is that it is *digit-additive*.

Definition 2.1. A sequence $h : \mathbb{N} \rightarrow \mathbb{Z}/m\mathbb{Z}$ is *k-digit-additive* if

$$h(k^L X + y) \equiv h(X) + h(y) \pmod{m} \quad \text{for all } X \geq 0, L \geq 0, 0 \leq y < k^L. \quad (1)$$

Equivalently, $h(i) = \sum_j w_{d_j} \bmod m$ is a weighted digit sum with weights $w : \Sigma_k \rightarrow \mathbb{Z}/m\mathbb{Z}$ satisfying $w_0 = 0$: the representation of $k^L X + y$ is that of X followed by the zero-padded length- L representation of y , and $w_0 = 0$ makes the padding harmless. Condition (1) forces $h(0) = 0$, and the associated DFAO $\delta(q, d) = q + w_d$ is zero-consistent. Here $t_{2,m}$ is the case $k = 2$, $(w_0, w_1) = (0, 1)$.

3. LEMMA A: NERODE CLASSES ARE EXACTLY THE WINDOWS

Lemma 3.1 (Lemma A). *Let $k \geq 2$, $m \geq 2$, let $h : \mathbb{N} \rightarrow \mathbb{Z}/m\mathbb{Z}$ be k -digit-additive, and let $n \geq 1$. For $j \geq 0$ put*

$$W(j) = (h(nj + g))_{0 \leq g < n} \in (\mathbb{Z}/m\mathbb{Z})^n.$$

Then two words $x, y \in \Sigma_k^$ are Myhill–Nerode equivalent for the sequence $(h(ni))_{i \geq 0}$ if and only if $W([x]_k) = W([y]_k)$. Consequently*

$$\text{sc}(h(ni)) = \#\{W(j) : j \geq 0\},$$

that is, the reachable part of the Theorem 9(a) automaton of [6] is already minimal.

Proof. Recall that $x \sim y$ means $h(n[xz]_k) = h(n[yz]_k)$ for every $z \in \Sigma_k^*$.

Sufficiency. Let $z \in \Sigma_k^L$, so $n[xz]_k = k^L \cdot n[x]_k + n[z]_k$. Write $n[z]_k = k^L g + w$ with $0 \leq w < k^L$ and $g = \lfloor n[z]_k / k^L \rfloor$; since $[z]_k \leq k^L - 1$ we have $n[z]_k < nk^L$, hence $g \leq n - 1$. Thus $n[xz]_k = k^L(n[x]_k + g) + w$, and (1) gives

$$h(n[xz]_k) = h(n[x]_k + g) + h(w) = W([x]_k)[g] + h(w) \pmod{m}. \quad (2)$$

The right-hand side depends on x only through $W([x]_k)$, so $W([x]_k) = W([y]_k)$ implies $x \sim y$.

Necessity. Suppose $W([x]_k)[g] \neq W([y]_k)[g]$ for some $0 \leq g < n$. Choose L with $k^L \geq n$ and set $\varphi(Z) = \lfloor nZ/k^L \rfloor$ for $0 \leq Z < k^L$. Then $\varphi(0) = 0$, $\varphi(k^L - 1) = \lfloor n - n/k^L \rfloor = n - 1$ (as $0 < n/k^L \leq 1$), and $\varphi(Z + 1) - \varphi(Z) \leq \lceil n/k^L \rceil = 1$, so φ is onto $\{0, \dots, n - 1\}$. Pick Z with $\varphi(Z) = g$ and let $z \in \Sigma_k^L$ be the zero-padded representation of Z . Applying (2) for both x and y , the $h(w)$ coincide and

$$h(n[xz]_k) - h(n[yz]_k) = W([x]_k)[g] - W([y]_k)[g] \not\equiv 0 \pmod{m},$$

so $x \not\sim y$.

Finally $W([x]_k)$ depends on x only through $[x]_k$, and every $j \geq 0$ equals $[x]_k$ for some x , so the Nerode classes biject with $\{W(j) : j \geq 0\}$. $\square$

Remark 3.2. Lemma 3.1 generalises [6, Thm. 16] (the case $k = m = 2$, n odd), and the proof is theirs. Two points are worth isolating: it holds for *every* n , even or odd, and it needs *no* reachability hypothesis. For digit-additive sequences the Theorem 9(a) construction is thus always optimal; the only remaining question is which windows occur at multiples of n .

4. LEMMA B: FOR ODD n , EVERY FACTOR OCCURS AT A MULTIPLE OF n

Lemma 4.1 (Lemma B). *Let $m \geq 2$, $h = t_{2,m}$, and let $n \geq 1$ be odd. Then every length- n factor of h occurs at some position divisible by n . Consequently, by Lemma 3.1,*

$$\text{sc}(t_{2,m}(ni)) = \rho_{t_{2,m}}(n).$$

Proof. Write $s = s_2$ and fix t with $2^t \geq n$. Every position P factors uniquely as $P = 2^t a + b$ with $0 \leq b < 2^t$. For $0 \leq g < n \leq 2^t$ there are two cases: if $b + g < 2^t$ then $h(P + g) = h(a) + h(b + g)$, while if $b + g \geq 2^t$ then $0 \leq b + g - 2^t < 2^t$ and $P + g = 2^t(a + 1) + (b + g - 2^t)$, so $h(P + g) = h(a + 1) + h(b + g - 2^t)$; both by digit-additivity. Hence the length- n factor at P is a function of the *type*

$$(u, v, b) = (h(a), h(a + 1), b).$$

It therefore suffices, given P of type (u, v, b) , to produce $j \geq 0$ such that nj has the same type.

Step 1: available types at multiples of n . As n is odd it is invertible modulo 2^t ; choose $j_0 \in [0, 2^t)$ with $nj_0 \equiv b \pmod{2^t}$. Then $nj_0 - b$ is a multiple of 2^t exceeding -2^t , so $a_0 := (nj_0 - b)/2^t \in \mathbb{N}$. For $j = j_0 + 2^t j'$, $j' \geq 0$,

$$nj = 2^t(a_0 + nj') + b,$$

so the high parts realisable with this b are exactly the integers $a' \geq a_0$ with $a' \equiv a_0 \pmod{n}$.

Step 2: hitting the pair (u, v) . Choose $T \in \{1, \dots, m\}$ with $T \equiv u - v + 1 \pmod{m}$, choose any $L > T$, and consider

$$a' = 2^L A + (2^T - 1), \quad A \geq 0.$$

Bits $0, \dots, T-1$ of a' are 1 and bit T is 0 (here $L > T$ is used), so $s(a') = s(A) + T$; and $a' + 1 = 2^L A + 2^T$, so $s(a' + 1) = s(A) + 1$, no carry reaching A . Hence if $h(A) \equiv v - 1 \pmod{m}$ then

$$h(a' + 1) = (v - 1) + 1 = v, \quad h(a') = (v - 1) + T \equiv u \pmod{m},$$

as required.

Step 3: the congruence. The constraint $a' \equiv a_0 \pmod{n}$ reads $2^L A \equiv a_0 - 2^T + 1 \pmod{n}$, and 2 is invertible modulo the odd number n , so it is a single congruence $A \equiv A_0 \pmod{n}$. By Gelfond's theorem [4] — for a base $q \geq 2$ and a modulus m with $\gcd(m, q-1) = 1$, the digit sum s_q is equidistributed modulo m along every arithmetic progression; for $q = 2$ the hypothesis reads $\gcd(m, 1) = 1$ and is automatic — there are infinitely many $A \equiv A_0 \pmod{n}$ with $s(A) \equiv v-1 \pmod{m}$. Choosing such an A large enough that $a' \geq a_0$ and setting $j' = (a' - a_0)/n \geq 0$, $j = j_0 + 2^t j'$, the multiple nj has type (u, v, b) and hence carries the same length- n factor as P .

By Lemma 3.1, $\text{sc}(t_{2,m}(ni)) = \#\{W(j) : j \geq 0\}$, and the $W(j)$ are exactly the length- n factors of $t_{2,m}$ at multiples of n ; by the above these are all of them. $\square$

Remark 4.2. Lemma 4.1 is the m -general replacement for [6, Lemma 15], which covers $m = 2$ by citing [2]; it re-proves [6, Thm. 14] without that citation. Oddness of n is used twice (invertibility of $n \pmod{2^t}$, of $2 \pmod{n}$) and cannot be dropped: for even n , $\#\{W(j)\} < \rho_{t_{2,m}}(n)$ strictly. This costs nothing, since $t_{2,m}(2i) = t_{2,m}(i)$ makes $\text{sc}(t_{2,m}(ni))$ depend only on the odd part of n .

5. THEOREM A

The third ingredient is the factor complexity of $t_{2,m}$, determined by Tromp and Shallit [8] and, for all bases, by Starosta [7], whose Table 2 gives, for the base- b word modulo m with q the order of $b-1$ in $\mathbb{Z}/m\mathbb{Z}$ (so $q = m$ for $b = 2$),

$$\begin{aligned} C(n) &= qm(n-1) - m(b^\kappa - b^{\kappa-1}) && \text{for } n = b^\kappa + 1 + \ell, \ 0 \leq \ell < b^\kappa - b^{\kappa-1}, \\ C(n) &= qm(n-1) - m(b^\kappa - b^{\kappa-1} + \ell) && \text{for } n = (2b-1)b^{\kappa-1} + 1 + \ell, \ 0 \leq \ell < b^{\kappa+1} - 2b^\kappa + b^{\kappa-1}. \end{aligned}$$

For $b = 2$ the two regimes merge.

Proposition 5.1. *For $n \geq 3$ let $p \geq 0$ and $0 < r \leq 2^p$ be the unique integers with $n-1 = 2^p + r$. Then $\rho_{t_{2,m}}(n) = m^2(n-1) - m \max(2^{p-1}, r)$, where 2^{p-1} is read as the real number $1/2$ when $p = 0$.*

The reduction is direct: the first regime is $r = \ell < 2^{\kappa-1}$ (with $\ell = 0$ giving $p = \kappa - 1$, $r = 2^{\kappa-1}$), the second $r = 2^{\kappa-1} + \ell \geq 2^{\kappa-1}$; at $m = 2$ this specialises to Brlek's formula [3], whence [6, Cor. 17]. The case $p = 0$ occurs only for $n = 3$, where $\max(1/2, 1) = 1$ and $\rho_{t_{2,m}}(3) = 2m^2 - m$; readers preferring integers may treat $n = 3$ as a base case.

Theorem 5.2 (Theorem A). *Let $m \geq 2$ and let $n \geq 3$ be odd. Write $n-1 = 2^p + r$ with $0 < r \leq 2^p$. Then*

$$\text{sc}(t_{2,m}(ni)) = \rho_{t_{2,m}}(n) = m^2(n-1) - m \max(2^{p-1}, r).$$

Consequently, for every odd $n \geq 3$,

$$\text{sc}(t_{2,m}(ni)) \geq \left(1 - \frac{1}{2m}\right)m^2(n-1) \geq \frac{3}{4}m^2(n-1) \geq \frac{1}{2}nm^2,$$

and along the odd values $n = 3 \cdot 2^{p-1} + 1$ ($p \geq 2$) the state count equals $(1 - \frac{1}{3m})m^2(n-1)$ exactly, so that

$$\limsup_{\substack{n \rightarrow \infty \\ n \text{ odd}}} \frac{\text{sc}(t_{2,m}(ni))}{nm^2} = 1 - \frac{1}{3m} \geq \frac{5}{6}.$$

In particular the family $(t_{2,m})_{m \geq 2}$ answers Problem 1 with $C = 1/2$ — for all odd $n \geq 3$, not merely infinitely many.

Proof. The first equality is Lemma 4.1 (which uses Lemma 3.1), the second is Proposition 5.1.

For the bounds put $N = n - 1 = 2^p + r$. Since $r > 0$, $2^p \leq N$ and so $2^{p-1} \leq N/2$; since $r \leq 2^p$, $N \geq 2r$ and so $r \leq N/2$. Hence $\max(2^{p-1}, r) \leq N/2$, with equality iff $r = 2^p$, and $\text{sc} \geq m^2 N - mN/2 = (1 - \frac{1}{2m})m^2 N \geq \frac{3}{4}m^2 N$ as $m \geq 2$; finally $N \geq \frac{2}{3}n$ for $n \geq 3$, giving $\text{sc} \geq \frac{1}{2}nm^2$.

For the limsup, minimise $f(p, r) = \max(2^{p-1}, r)/(2^p + r)$ over $0 < r \leq 2^p$: on $r \leq 2^{p-1}$ it is $2^{p-1}/(2^p + r)$, decreasing in r , and on $r \geq 2^{p-1}$ it is $r/(2^p + r)$, increasing in r , so the minimum is $f = 1/3$ at $r = 2^{p-1}$. Thus $\max(2^{p-1}, r) \geq (n-1)/3$ always, with equality exactly when $n = 3 \cdot 2^{p-1} + 1$ (odd precisely for $p \geq 2$). Hence $\text{sc}(t_{2,m}(ni)) \leq (1 - \frac{1}{3m})m^2(n-1) < (1 - \frac{1}{3m})nm^2$ for every odd $n \geq 3$, with equality in the first inequality along $n = 3 \cdot 2^{p-1} + 1$, where $(n-1)/n \rightarrow 1$; and $1 - 1/(3m) \geq 5/6$ for $m \geq 2$. $\square$

Remark 5.3. Against the bound actually proved in [6], namely $2nm^2$ for $k = 2$, the family realises the fraction $(1 - \frac{1}{3m})/2 \in [\frac{5}{12}, \frac{1}{2}]$: Corollary 10(a) is tight in the shape nm^2 , loose only in the factor k (cf. Theorem 7.1).

6. VERIFICATION

Theorem 5.2 is a proof, conditional only on the two literature inputs (Proposition 5.1 and Gelfond's theorem). It was nevertheless checked numerically by implementations sharing no code. Two computed sc directly: a *theory-free* msd Myhill–Nerode search, in which the signature of X is the raw table $(u(k^L X + j))_{L \leq L_{\max}, j < k^L}$ built from the oracle $u(i) = s_2(ni) \bmod m$ alone, with transitions $X \mapsto kX + d$ (distinct signatures are provably distinct states, so it cannot over-count); and a breadth-first construction of the Theorem 9(a) window automaton followed by Moore minimisation. Separately, $\rho_{t_{2,m}}(n)$ was computed by suffix doubling on a 2^{22} -symbol prefix, and $\#\{W(j)\}$ straight from population counts, with no automaton involved. A further independent code path is Peanut, which compiles the predicate $\mathbf{T[n*i]=a}$ into a minimal DFA.

Check	Scope	Result
Nerode search vs. window construction vs. Thm. 5.2	$m = 2 \dots 7$, odd $n = 3 \dots 101$ (300 cells)	no mismatch
Thm. 5.2 vs. window construction, adversarial parameters	$m \in \{8, 10, 12, 16, 20\} \times 7$ values of n ; 12 further cells up to $m = 32$	no mismatch
$\rho_{t_{2,m}}$ by suffix doubling vs. Prop. 5.1	$m = 2 \dots 7$, every n from 3 to 130	no mismatch, no non-convergence
$\#\{W(j) : j < J\}$ from population counts vs. $\rho_{t_{2,m}}(n)$ (Lem. 4.1)	slowest-converging cells, $J < 2 \cdot 10^7$	exact, e.g. $(m, n) = (7, 101) \mapsto 4648$
Peanut, $\mathbf{dfa} \ \mathbf{T[n*i]=a}$	14 cells, $m \leq 7$, $n \leq 101$	agrees with both constructions
subsequence $n = 3 \cdot 2^{p-1} + 1$, $p = 2 \dots 6$	$m \in \{2, 3, 5, 7\}$ (20 cells)	$\text{sc} = (1 - \frac{1}{3m})m^2(n-1)$ exactly
extremes of $\text{sc}/(nm^2)$, $\text{sc}/(m^2(n-1))$	the 300 cells above	minima $1/2$ (at $m = 2, n = 3$) and $3/4$

Two caveats about status. First, reading the state count of the per-letter DFA $\mathbf{dfa} \ \mathbf{T[n*i]=a}$ as sc presumes each level set $\{i : t_{2,m}(ni) = a\}$ needs the full sc states. This held in every cell run — all m letters gave the same count, equal to sc — but it is an empirical bridge, not a proof, so the engine figures corroborate the two constructions rather than replace them. Second, and more importantly:

Observation 6.1 (base 3; machine-verified only). Let $g_m(i) = (\#\{\text{digits equal to 2 in } (i)_3\}) \bmod m$, a 3-digit-additive sequence with weights $(0, 0, 1)$, generated by a minimal m -state zero-consistent base-3 DFAO. Then $\text{sc}(g_m(ni)) = \rho_{g_m}(n) = m^2(n-1)$ exactly whenever $3 \nmid n$, and $\text{sc}(g_m(ni)) = \text{sc}(g_m(n'i))$ for $n = 3^v n'$ with $3 \nmid n'$.

Observation 6.1 was reproduced independently for $m = 2 \dots 6$ and $n \in \{2, 4, 5, 7, 8, 11, 13, 16, 22, 31, 40, 44\}$, and is *not* proved here. Lemma 3.1 applies verbatim, so $\text{sc}(g_m(ni)) = \#\{W(j)\}$; the missing ingredients are an analogue of Lemma 4.1 — Gelfond's hypothesis $\gcd(m, q-1) = 1$ fails for $q = 3$ and even m , though the sequence here is a weighted digit sum rather than s_3 — and a proof that $\rho_{g_m}(n) = m^2(n-1)$. This family saturates nm^2 , reaching $\text{sc}/(nm^2) = 1 - 1/n$, but only one third of the knm^2 of Corollary 10(a). It is recorded as a conjecture, not a result.

7. REMARKS

A sharpening of Corollary 10(a). The window parametrisation of Lemma 4.1 is not special to $t_{2,m}$, and gives the following, in which the factor k of Corollary 10(a) becomes $1 + (k-1)/m$.

Theorem 7.1 (Theorem C). *Let h be generated by an m -state zero-consistent msd-first DFAO with interior sequence h' and intrinsic DFAO $\bar{M}$, let $n \geq 1$, and let t be minimal with $k^t \geq n$. Then*

$$\rho_{h'}(n) \leq \rho_{\bar{M}}(2)(n-1) + m(k^t - n + 1) \leq m^2(n-1) + m((k-1)n + 1),$$

and hence $\text{sc}(h(ni + c)) \leq m^2(n-1) + m((k-1)n + 1)$ for all $0 \leq c < n$.

Proof. Write $P = k^t a + b$, $0 \leq b < k^t$. Every position in $[P, P + n]$ is $k^t(a + e) + r$ with $e \in \{0, 1\}$, $0 \leq r < k^t$, and $h'(k^t A + r) = \delta(h'(A), r_1 \cdots r_t)$ where $r_1 \cdots r_t$ is r zero-padded to length t , since msd reading concatenates digit blocks — exactly where zero-consistency is needed. So the factor at P is determined by $(h'(a), h'(a + 1), b)$, and by $(h'(a), b)$ alone when $b + n \leq k^t$. There are $k^t - n + 1$ values of b of the latter kind, each contributing at most m factors, and $n - 1$ of the former, each contributing at most the number $\rho_{\widehat{M}}(2)$ of realised pairs $(h'(a), h'(a + 1))$. The second inequality uses $\rho_{\widehat{M}}(2) \leq m^2$ and $k^t \leq k(n - 1) \leq kn - 1$, the latter by minimality of t . $\square$

Zero-consistency is a genuine hypothesis, not a normalisation: without it $h'(0) = q_0$ fails, the recursion $h'(ki + b) = \delta(h'(i), b)$ breaks at $i = 0$, and the window construction of [6, Thm. 9] — hence Theorem 7.1 and every window count in this note — is invalid. Both families here are zero-consistent, as is every digit-additive sequence. Theorem 7.1 improves knm^2 when $m \gg k$, but it is a one-shot form of the standard blocking argument behind [1, Thm. 10.3.1] and should be regarded as folklore-grade.

On significance. This note assembles three known pieces rather than proving a new hard theorem. The proof of [6, Thm. 16] generalises essentially verbatim to Lemma 3.1; their Lemma 15 needs a replacement, supplied by the short construction of Lemma 4.1 with Gelfond’s 1968 theorem; and the growth rate $m^2 n$ is already in [8, 7]. What was absent from the source is precisely the join of the three. The answer appears correct and complete, but should be read as a routine composition, and may well be known to specialists in a form we have not located.

The constant is not improvable uniformly. Within this family $C = 1/2$ is exactly attained, at $(m, n) = (2, 3)$, where $\text{sc} = 6 = \frac{1}{2} \cdot 3 \cdot 2^2$. So $1/2$ is the largest constant valid for *all* odd $n \geq 3$ and all $m \geq 2$ here, and the stronger $1 - 1/(3m)$ of Theorem 5.2 is a statement about a subsequence of n only. Since Problem 1 asks only for infinitely many n , either constant answers it.

What is not settled. Three questions remain open. (i) Observation 6.1: the identity $\rho_{g_m}(n) = m^2(n - 1)$ and an analogue of Lemma 4.1 in base 3. (ii) The exact maximum of $\text{sc}(h(ni))$ over all m -state k -DFAOs. The bound $m^2(n - 1)$ holds exhaustively for all zero-consistent 2- and 3-state DFAOs at $k \in \{2, 3\}$, $n \in \{7, 11, 16, 31\}$, but fails for $k = 4$: weights $(0, 0, 1, 0)$ modulo 3 give $\text{sc}(h(31i)) = 312 > 270 = m^2(n - 1)$, still below Theorem 7.1’s bound of 372. The truth thus lies between $m^2(n - 1)$ and $m^2(n - 1) + m((k - 1)n + 1)$, and the factor k in knm^2 is a real but bounded effect that vanishes as m grows. (iii) The case $c > 0$, i.e. $(h(ni + c))_{i \geq 0}$ and [6, Cor. 10(b)]: only $c = 0$ is treated here, that being what Problem 1 asks.

Acknowledgement and declaration of AI usage. This note, and the work behind it, were produced with substantial use of a large language model, and the first version circulated (17 August 2026) failed to say so; the author apologises for that omission. Precisely: the models were Claude (Anthropic): `claude-fable-5` as the orchestrating model, with `claude-opus-5` and `claude-sonnet-5` sub-agents for search, implementation, refereeing and drafting, all used through the Claude Code agent framework; they were used for (i) the literature and prior-art search; (ii) the design and implementation of Peanut, the open-source decision procedure with which every computation here was run; (iii) the search for the family, the formulation of Lemmas 3.1–4.1 and Theorem 5.2, and their proofs; (iv) an adversarial “referee” pass, also carried out by the model, which re-derived each step, wrote independent brute-force checks sharing no code with Peanut, and tried to construct counterexamples; and (v) the drafting of this text, most of which is the model’s output lightly edited by the author. The author directed the work, chose the problem, read and takes responsibility for every statement, and ran the computations. All finite claims were verified by the independent implementations described in Section 6; the referee transcripts and code are public in the Peanut repository. Any errors are the author’s.

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